



Adaptive Slot-ahead Control for nFAP I Split under Unpredictable network congestion

應用於 nFAP I 拆分架構在不可預測網路壅塞下之自
適應提前時槽控制機制

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OUTLINE



PART 1 Introduction



PART 2 System Architecture



PART 3 Proposed Method



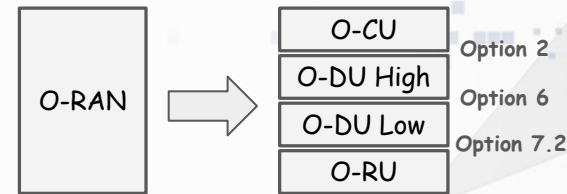
PART 4 Experimental Results



PART 5 Conclusion



5G O-RAN Split Options

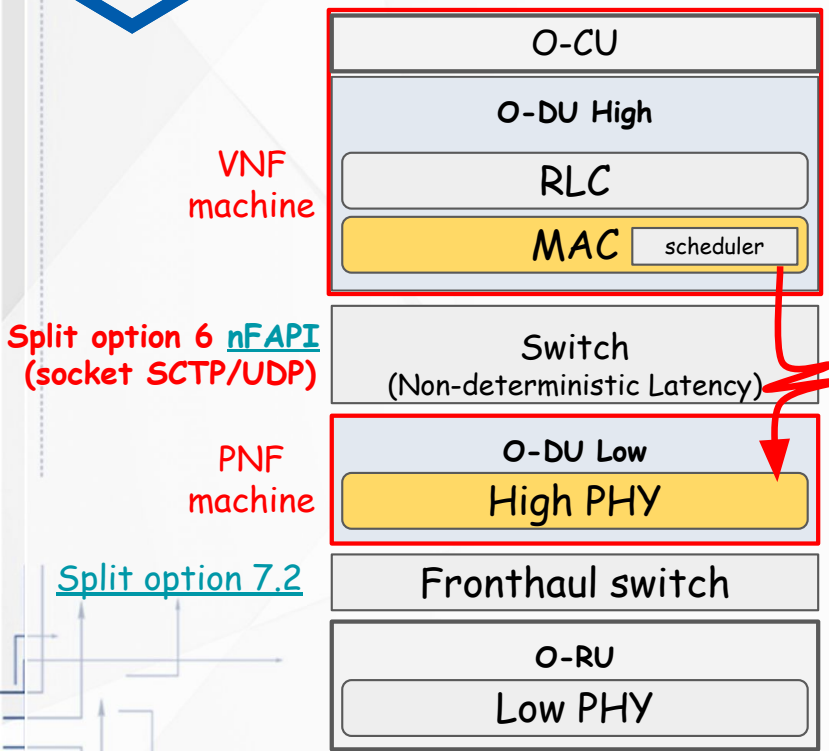


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Split Option	Sync level	Core Management	Deployment Environment
Option 2 (O-CU/O-DU Split)	Sequence Number	Flow control & Deep Buffering	General packet-switched networks
Option 6 nFAPI (O-DU High/Low split)	Frame/ Slot	Delay management	General packet-switched networks
Option 7.2 (O-DU/O-RU Split)	Phase/ Symbol	Hardware PTP	Time-Sensitive Networking



nFAPI: network Functional API



Advantages

- Disaggregated in different machine (Multi-vendor, C-RAN)
- Standard: [SCF225 explicitly supports](#) Non-ideal Fronthaul to enable low-cost deployments.

Challenges

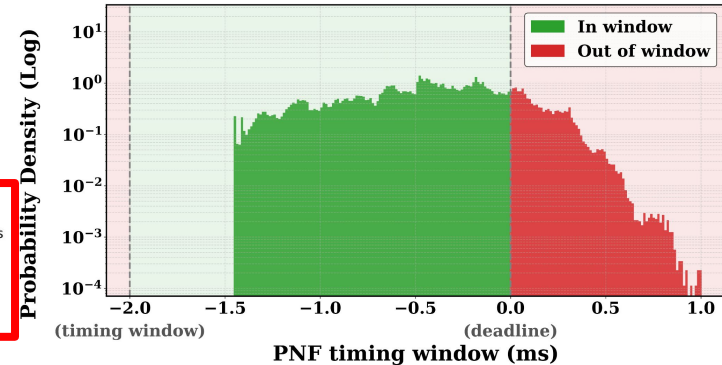
- Slot-level synchronization is mandatory.
- Variable VNF-PNF latency under different traffic loads.
- Unpredictable latency from the non-ideal environment.



DL_TTI: Downlink Transmission Time Interval



- **VNF Slots Ahead:**
VNF schedules and transmits packets in advance.
- **PNF Window:**
Strictly enforced reception window. Packets outside this window are discarded.

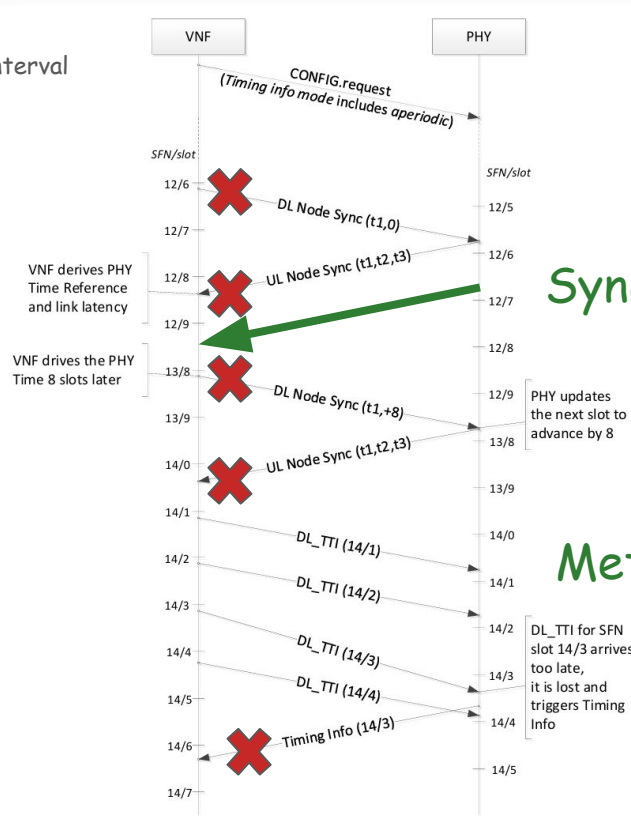




OAI Implement nFAPI

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DL_TTI: Downlink Transmission Time Interval



Sync: Slot.indication

Methods: Static Slots Ahead



Exponentially Weighted Moving Average (EWMA)

$\text{Difference} = \text{SampleRTT} - \text{EstimatedRTT}$

$\text{EstimatedRTT} = (1 - \alpha) \times \text{EstimatedRTT} + \alpha \times \text{SampleRTT} \quad (\alpha = 1/8)$

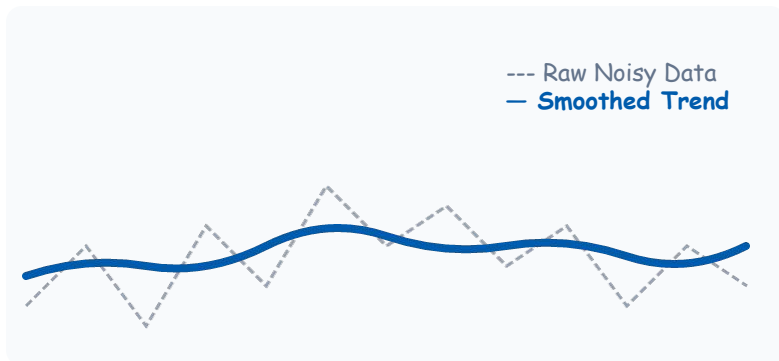
$\text{Deviation} = (1 - \beta) \times \text{Deviation} + \beta \times |\text{Difference}| \quad (\beta = 1/4)$

$\text{TimeOut (RTO)} = \text{EstimatedRTT} + 4 \times \text{Deviation}$



Exponentially Weighted Moving Average (EWMA)

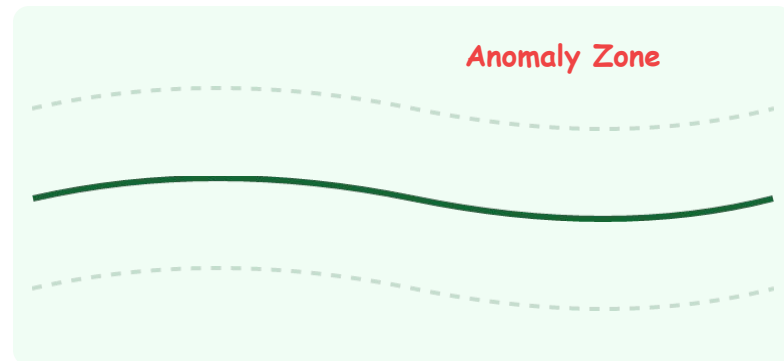
1. Data Smoothing (EWMA)



$$St = \alpha \cdot Xt + (1-\alpha) \cdot St-1$$

Filters high-frequency noise

2. Anomaly Detection



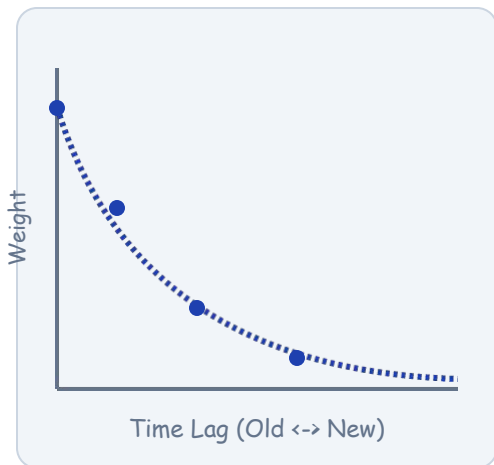
$$St > Est \pm 3 \times Dev$$

Anomaly detected when smoothed trend (St) exceeds the confidence interval



Exponentially Weighted Moving Average (EWMA)

EWMA: Weight Distribution Over Time



Core Logic

Recursive Smoothing (The Formula)

$$S_t = \alpha X_t + (1 - \alpha) S_{t-1}$$

Relationship to Exponential Decay

- The recursive expansion results in a weighted sum of all past data.
- Weights decrease geometrically as $(1 - \alpha)^k$, favoring recent points.
- Provides noise reduction while maintaining responsiveness.



Related Works (paper):

E2E: End-to-end

HARQ: Hybrid Automatic Repeat Request

Paper	Core Strategy	Congestion Handling	Evaluation Metrics	Level of Validation
[1]	Queuing Model	X (Predict only)	Latency	C++ Simulator
[2]	HARQ Pre-allocation	Reactive retransmissions	Throughput	MATLAB Simulator
This thesis	Delay management	Proactive tracking	Throughput, Latency	End-to-end

reference:

[1] [L. Díez, A. M. Alba, W. Kellerer and R. Agüero, "Flexible Functional Split and Fronthaul Delay: A Queuing-Based Model," in IEEE Access, vol. 9, pp. 151049-151066, 2021](#)

[2] [M. Elfiky, Z. Becvar and P. Mach, "Allocation of Resources for HARQ Retransmission in Mobile Networks Based on C-RAN," in IEEE Systems Journal, vol. 18, no. 1, pp. 450-461, March 2024](#)



OAI: OpenAirInterface
SCF: Small Cell Forum

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Problem: Distributed Deployment

- Unpredictable network congestion necessitates slot synchronization and advance scheduling.

Challenges: Static Scheduling Rigidity

- Static slot-ahead fail to adapt to network congestion.

Contributions: Delay Management Algorithm

- Adaptive slot-ahead control based on unpredictable network congestion.

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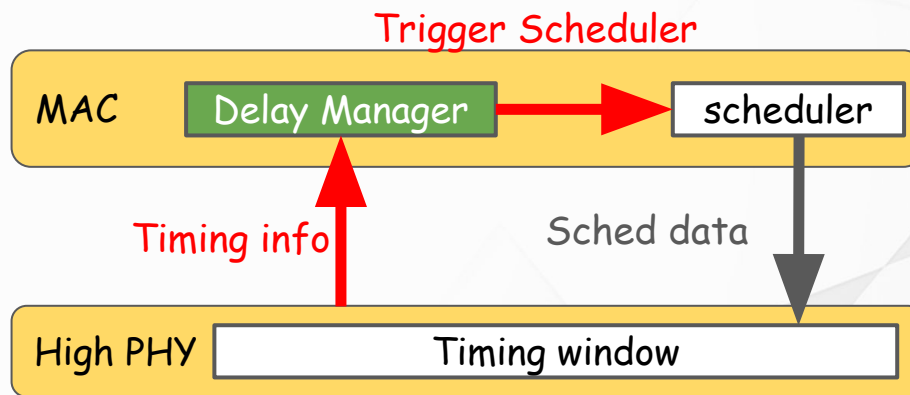
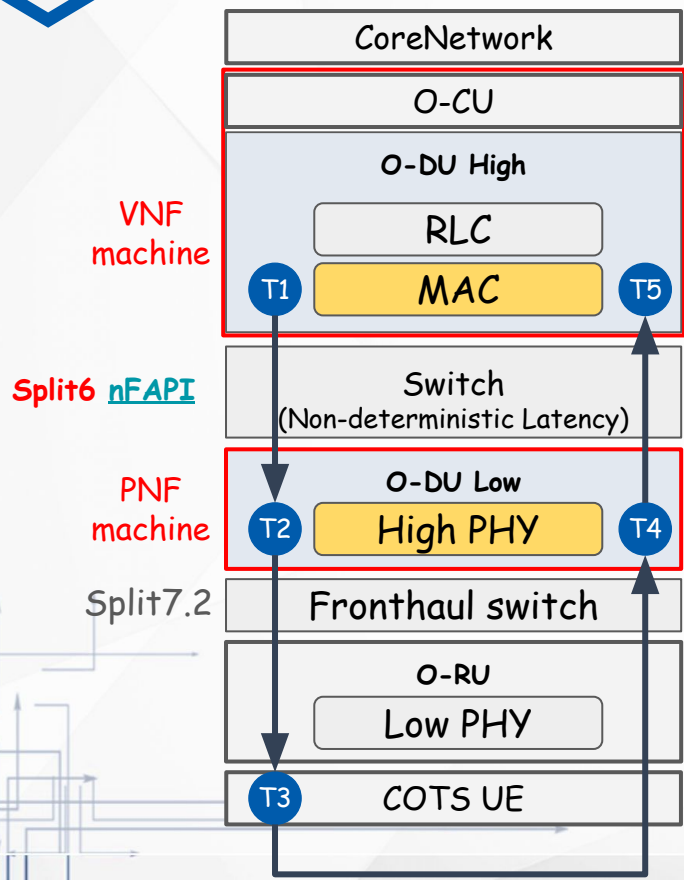


PART 5 Conclusion



System Architecture

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Key Validation Metrics

1. **MAC Layer HARQ RTT:** $T5 - T1$
2. **VNF-PNF Latency:** $T2 - T1, T5 - T4$

Note: RTT = Round-Trip Time

HARQ = Hybrid Automatic Repeat Request

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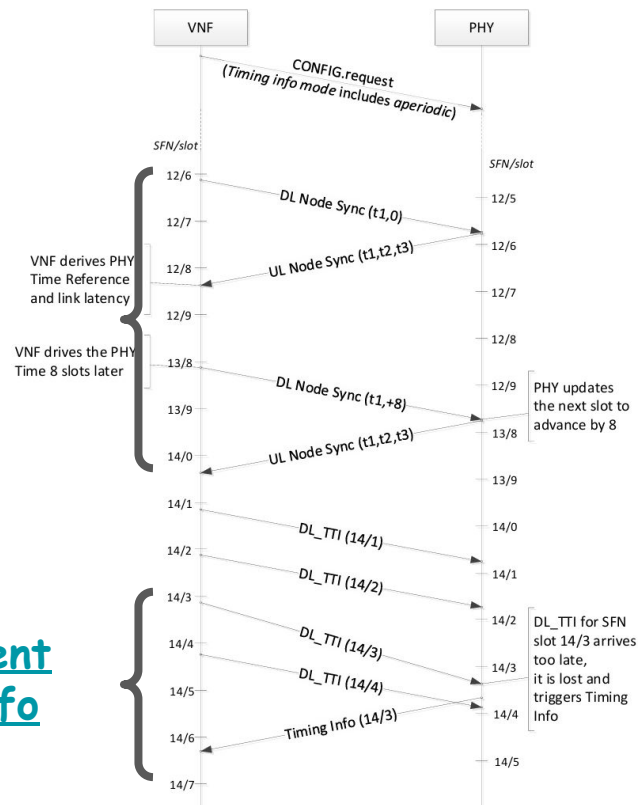


Delay management timing sequence

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DL_TTI: Downlink Transmission Time Interval

Node Sync



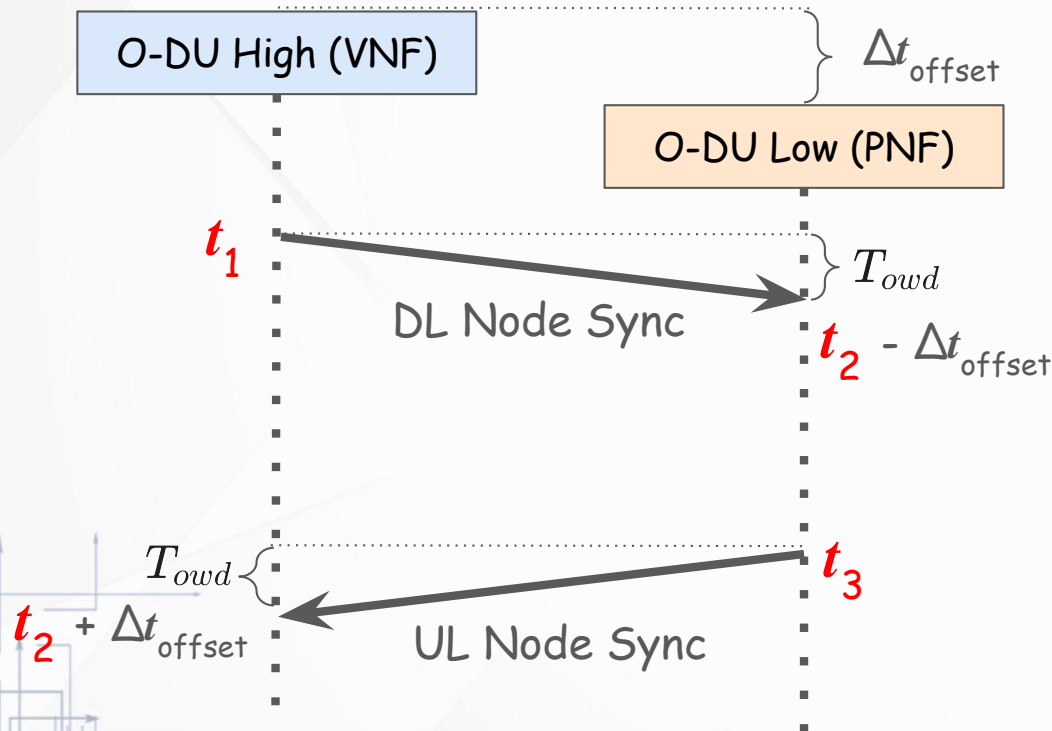
Delay management handle timing info



Delay Management: Node Sync

T_{owd} : one-way-delay

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Input: $t_1 t_2 t_3 t_4$
Output: Δt_{offset}

$$T_{owd} = (t_2 - \Delta t_{offset}) - t_1$$

$$T_{owd} = t_4 - (t_3 + \Delta t_{offset})$$

$$\Delta t_{offset} = \frac{(t_2 - t_1) - (t_4 - t_3)}{2}$$

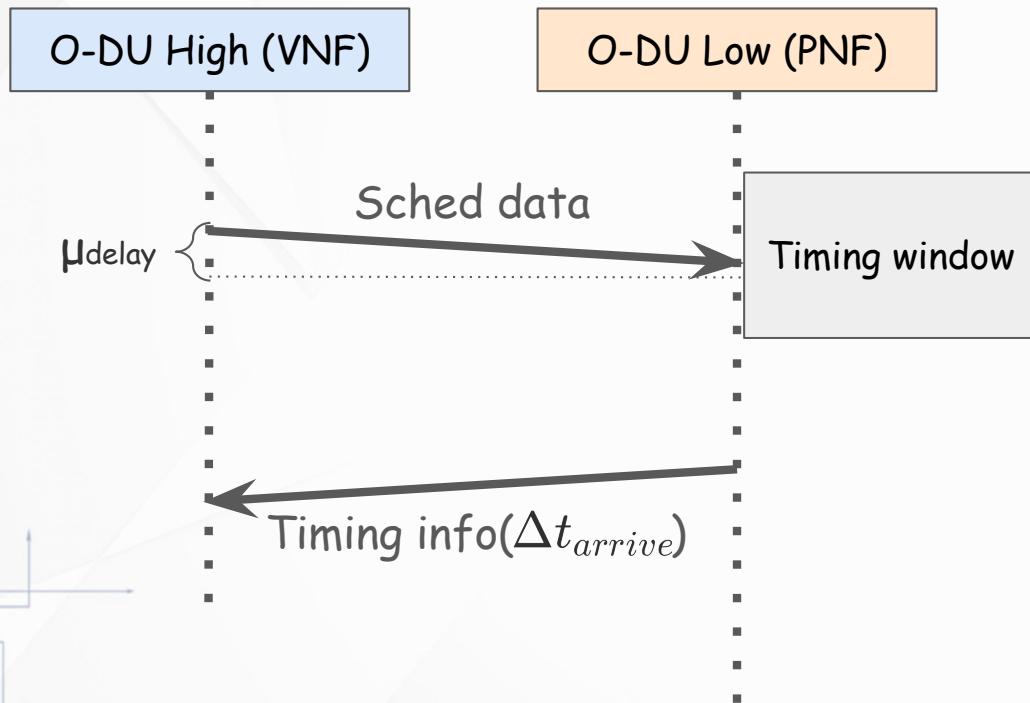


Delay Management

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Δt_{arrive} : VNF sched data arrive

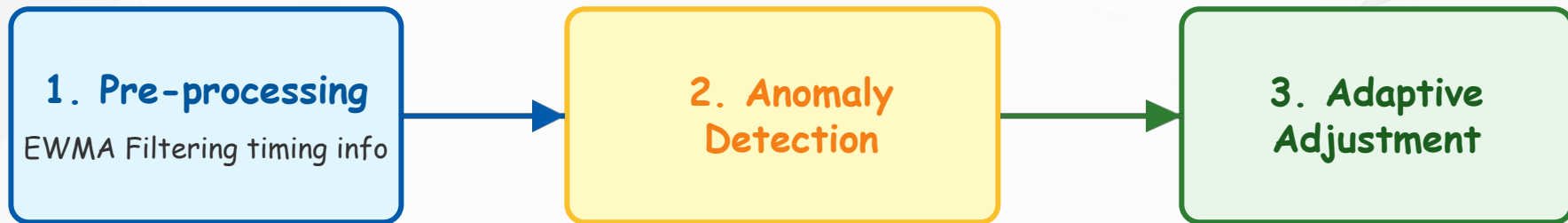
Input: **Timing info**
Output: **Scheduler timing**





Delay Management

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 $\text{dev}_{\text{delay}}$: EWMA jitter μ_{delay} : EWMA latency $\text{dev}_{\text{delay}}, \mu_{\text{delay}}$ 

EWMA Filtering avoids directly use the timing info reported for adjustments

$$|\Delta t_{\text{arrive}} - \mu_{\text{delay}}| > 3 * \text{dev}_{\text{delay}}$$

Adaptive Timing Adjustment: Anomaly: Increase slot ahead / Stable: Decrease slot ahead (Optimize RTT)



Auto-testing rApp

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Deployment & Control

- Automated gNB E2E Startup
- ADB Control for COTS UE

Testing & Analysis

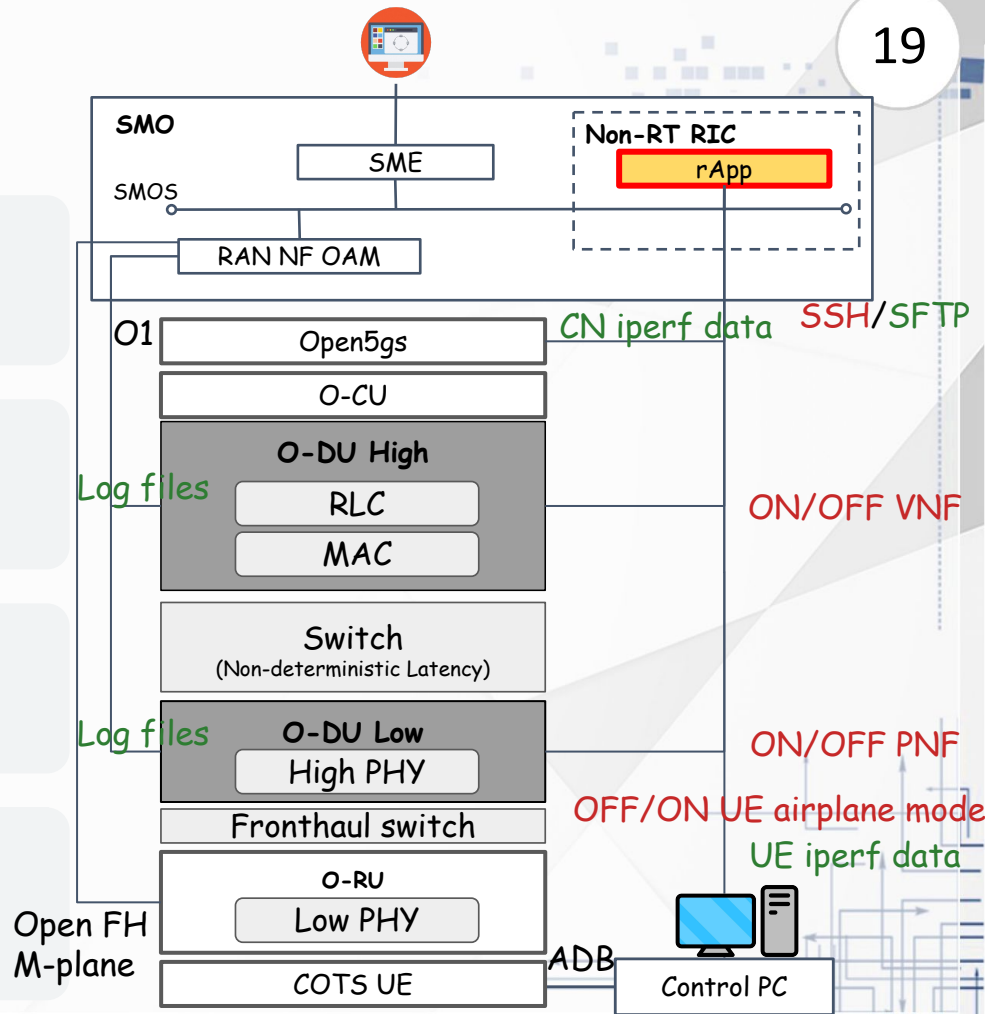
- iPerf and Ping Testing
- Analytical Chart Generation

Data Management

- Automated Data Collection (gNB logs, iperf)
- File Backup to GitHub

Automated Self-Healing

- RU M-plane / Reboot
- NIC Reset



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Overview of Experiments

Experiments were conducted to verify the proposed delay management method in an **OAI nFAPI End-to-End** environment. Each reported throughput value represents the average of 180 data points (3 independent trials $\times$ 60 samples per trial).

- **Experimental 1:**
Validation of EWMA for PNF Timing Info
- **Experimental 2:**
Benchmark throughput & MAC RTT efficiency against static baselines.
- **Experimental 3:**
Stability under Network Congestion
- **Demo:**
Confirm end-to-end functional deployment through a rApp demonstration video.



Experiment & Validation

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Experimental Scenario

- Exp 1: Validation of EWMA for PNF Timing Info
- Exp 2: Performance: Throughput & MAC RTT Efficiency
- Exp 3: Stability under Network Congestion
- Demo rApp video



Experimental Scenario

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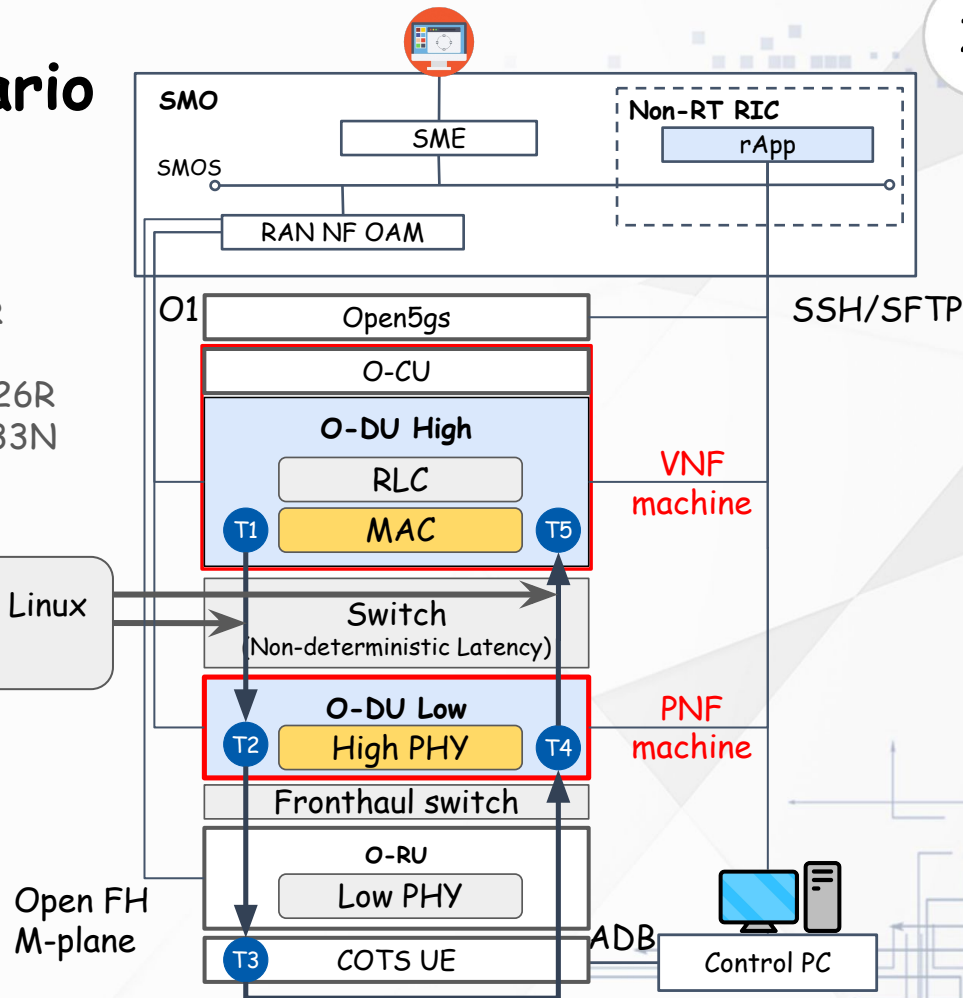
- Software
 - OAI (VNF,PNF) branch: 2026.w13
 - 30kHz, 3D1S1U, 4 Layer MIMO, 4T4R
- Hardware
 - VNF server: Intel(R) Xeon(R) Gold 6226R
 - PNF server: Intel(R) Xeon(R) Gold 6433N

Network latency simulation at T5-T4 and T2-T1 via Linux TC to validate split-architecture stability.

Key Validation Metrics

- MAC Layer HARQ RTT: T5 - T1
- VNF-PNF DL Latency: T2 - T1

Note: TC=Traffic Control





Experiment & Validation

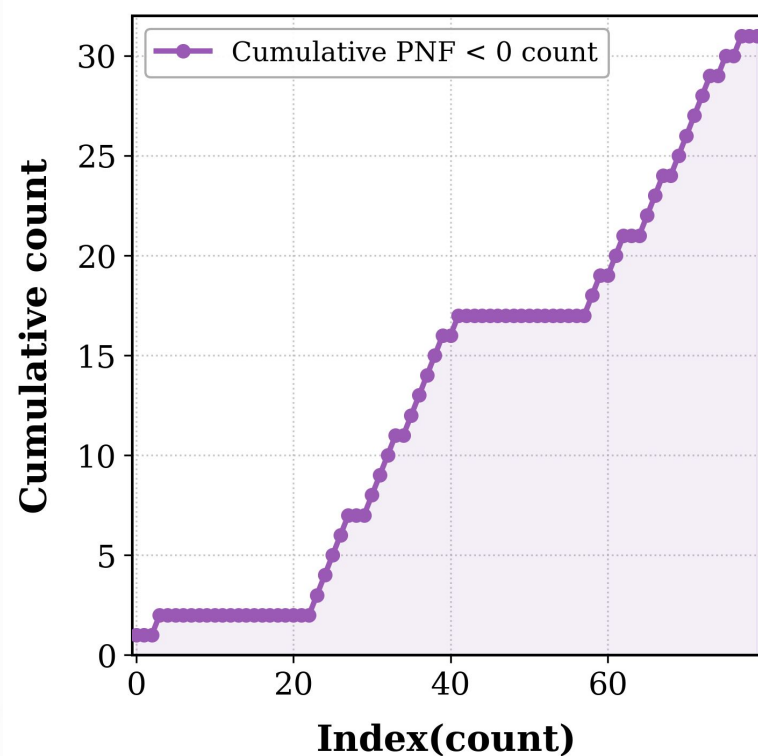
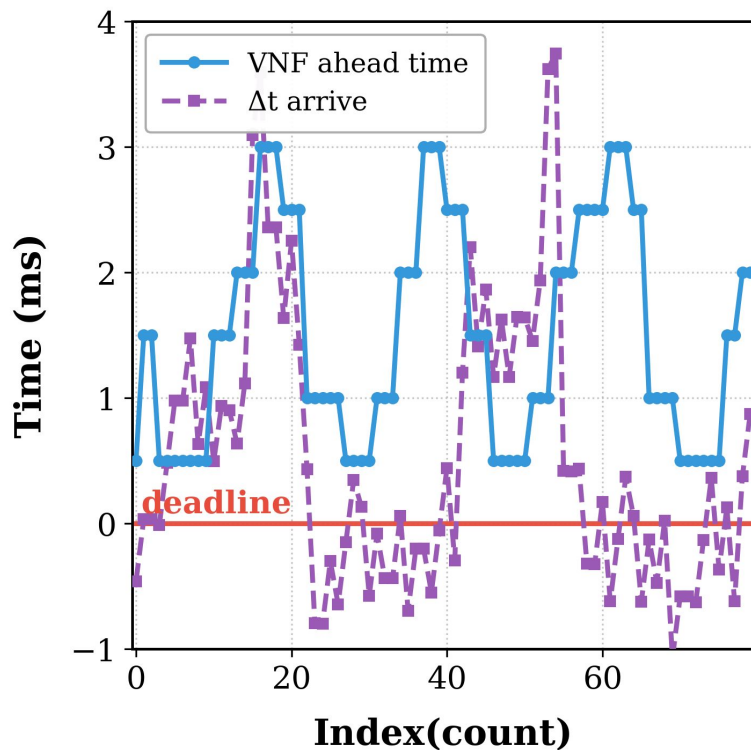
24

Experimental Scenario

- Exp 1: Validation of EWMA for PNF Timing Info
- Exp 2: Performance: Throughput & MAC RTT Efficiency
- Exp 3: Stability under Network Congestion
- Demo rApp video



Exp 1: Non-applied EWMA for PNF Timing Info

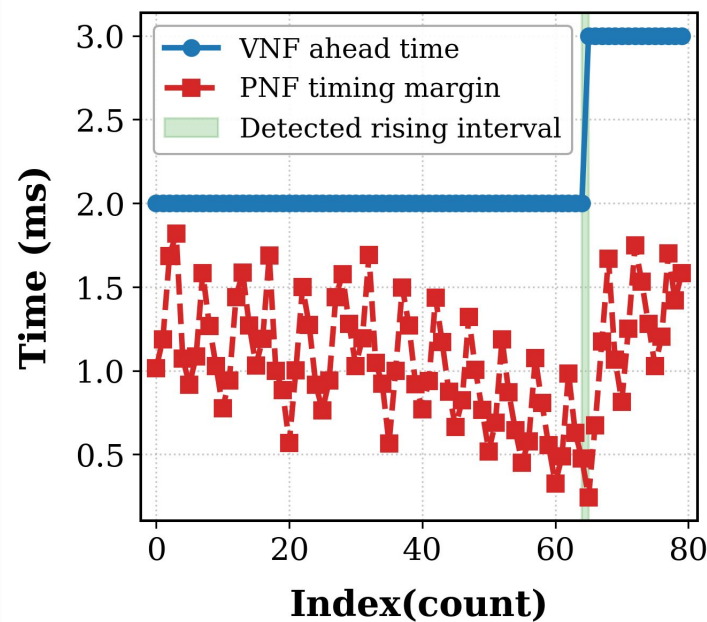




Exp 1: Applied EWMA for PNF Timing Info

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EWMA





Experiment & Validation

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Experimental Scenario

- Exp 1: Validation of EWMA for PNF Timing Info
- Exp 2: Performance: Throughput & MAC RTT Efficiency
- Exp 3: Stability under Network Congestion
- Demo rApp video

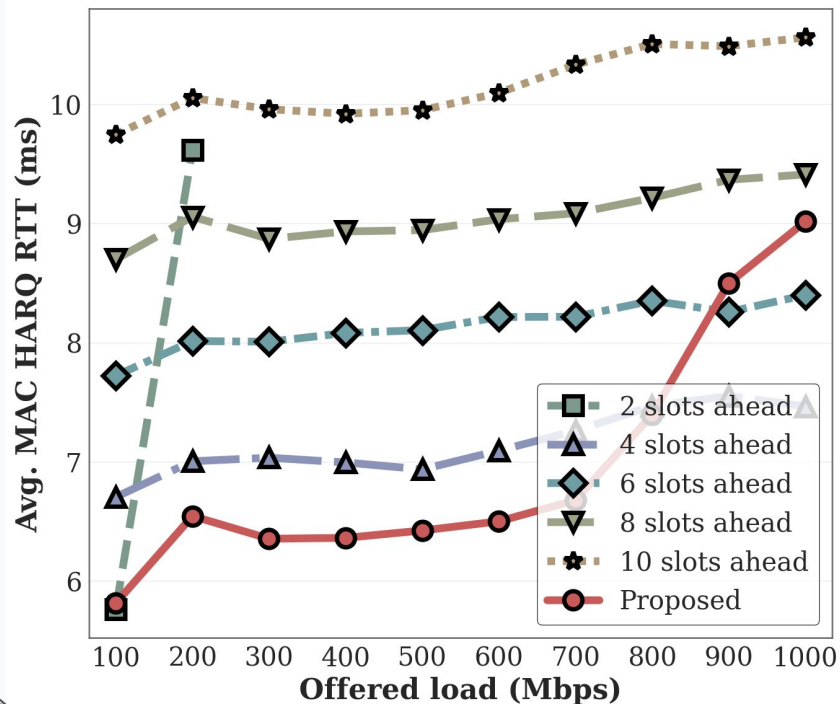


Exp 2: Performance: MAC RTT Efficiency & Throughput

Proposed Benefits

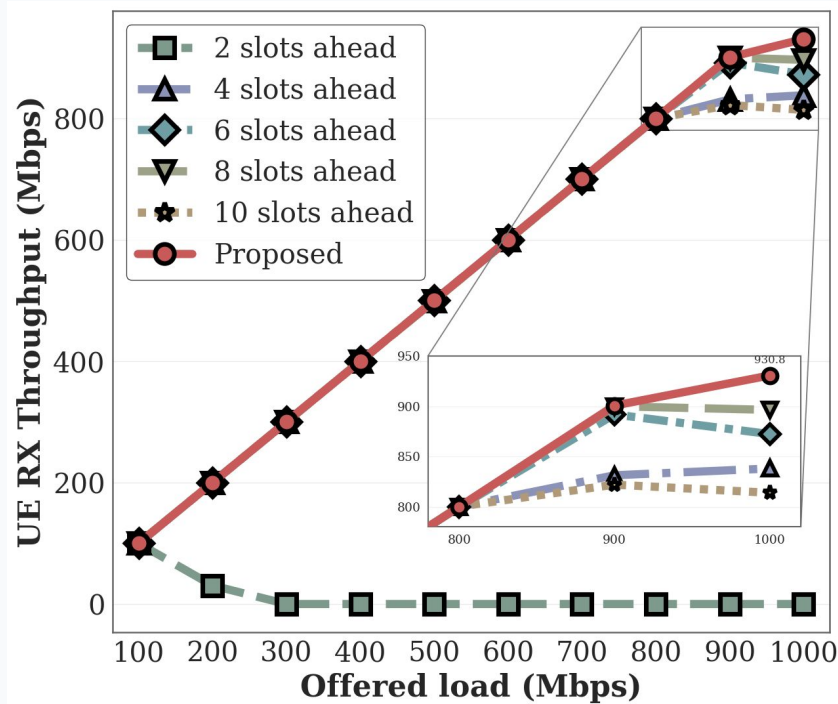
MAC Latency & Reliability

Better RTT under different Offered loads while ensuring minimal loss.



Throughput Optimization

The **best throughput** performance at peak loading.





Experiment & Validation

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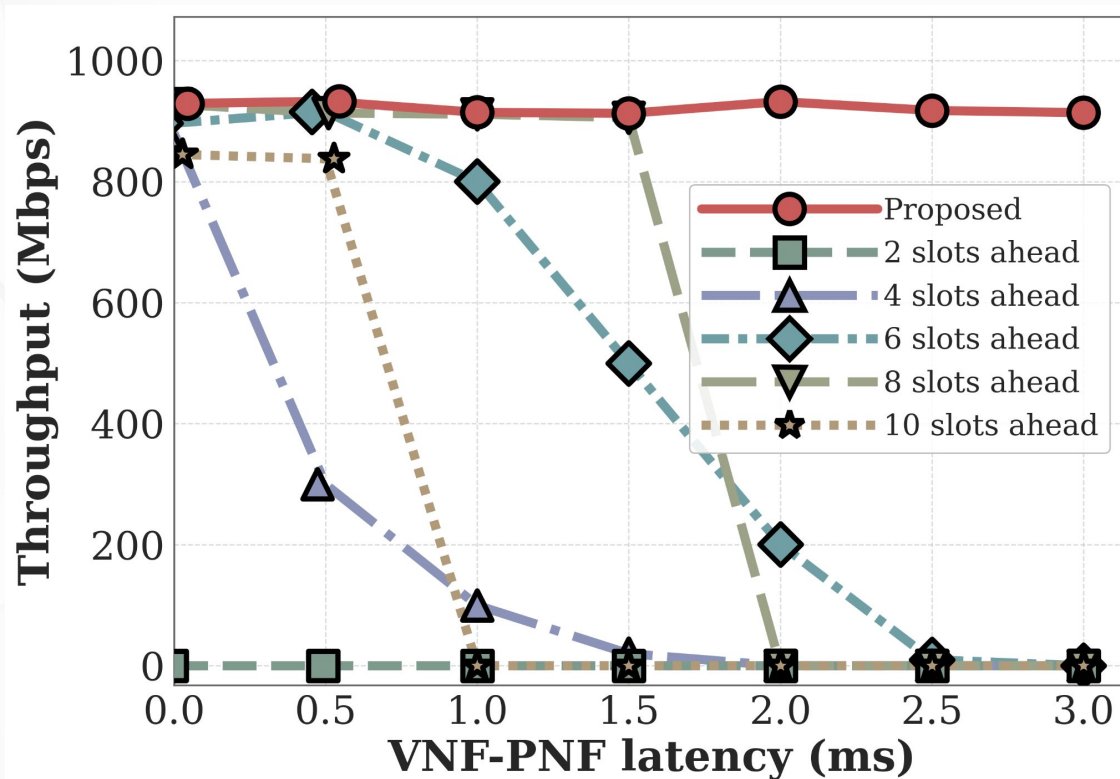
Experimental Scenario

- Exp 1: Validation of EWMA for PNF Timing Info
- Exp 2: Performance Benchmarking: Throughput & MAC RTT Efficiency
- **Exp 3: Stability under Network Congestion**
- Demo rApp video



Exp 2: Stability under Network Congestion

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Proposed method adapts to any latency to keep both connection and throughput stable.



Experiment & Validation

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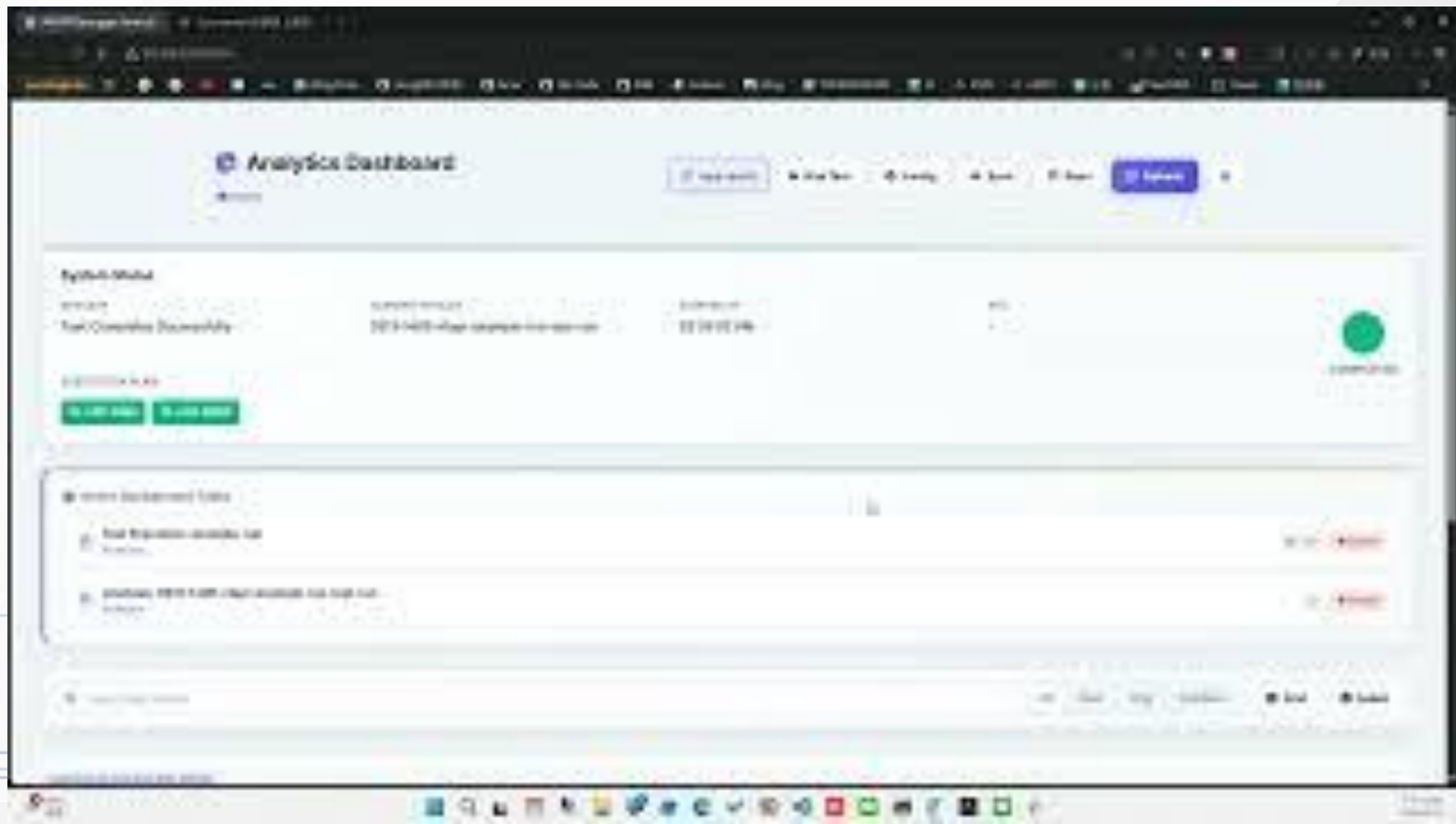
Experimental Scenario

- Exp 1: Validation of EWMA for PNF Timing Info
- Exp 2: Performance Benchmarking: Throughput & MAC RTT Efficiency
- Exp 3: Stability Assessment under Network Latency
- Demo rApp video



Demo video

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Conclusion

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⚠ The Problem

1. Adaptive Advance-Time Control

- Jacobson/Karels EWMA
- Maximum Advance Pinning
- PID Anti-Windup
- BBR-style Asymmetry

⚡ VNF Mechanisms

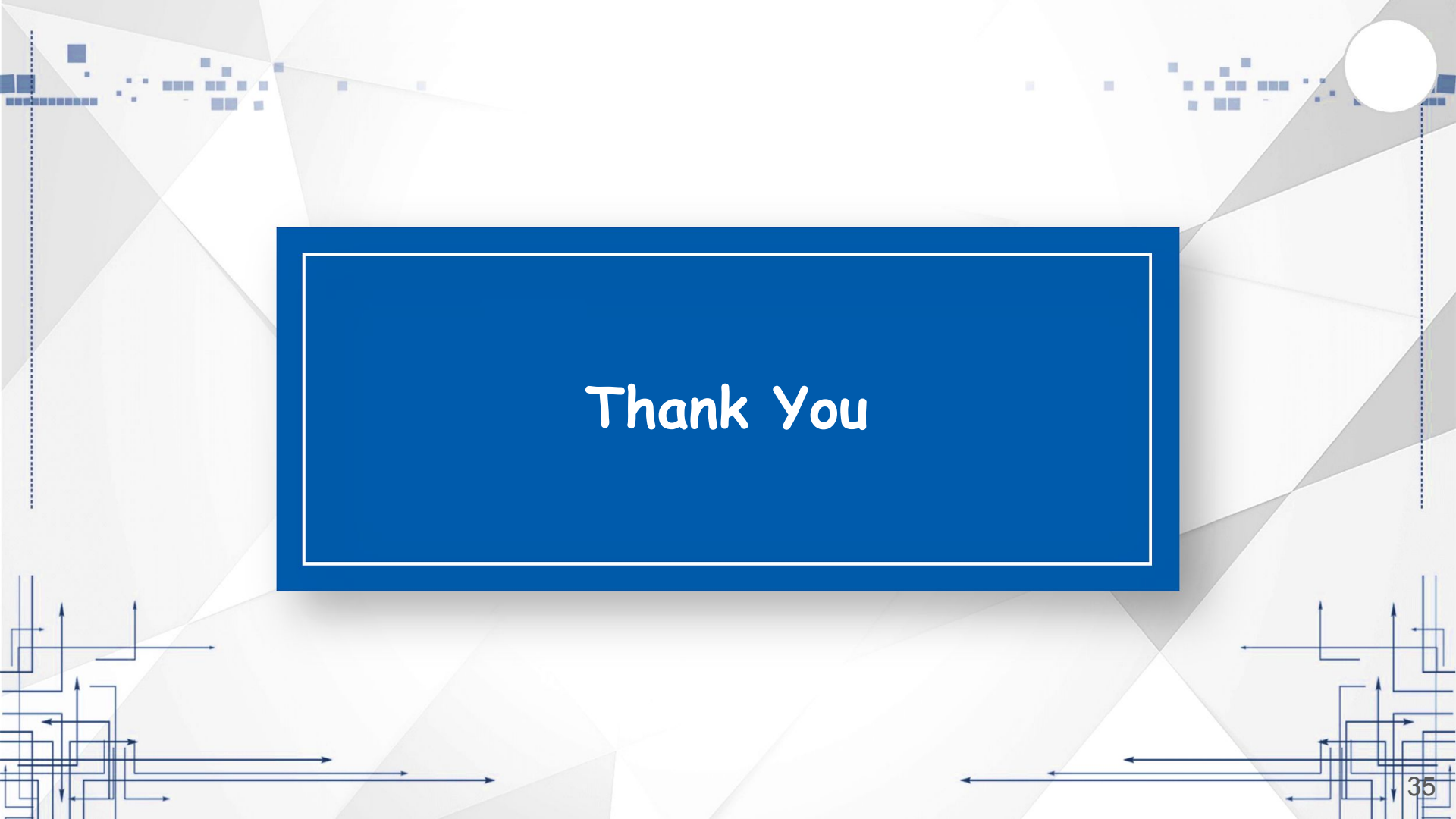
2. Autonomous VNF Timing

- Dedicated Thread
- High-Precision Sleep
- Burst Catch-up
- Extreme Drift Jump

↘ Performance Results

Implementation Benefits

- SCF Compliance
- Resilience
- Latency Reduction
- Stability



Thank You



Related Works (paper):

E2E: End-to-end

EMANE: Extendable Mobile Ad-hoc Network Emulator

HARQ: Hybrid Automatic Repeat Request

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Paper	Environment	Interface	spec. Compliance	Verify	E2E	Proposed Methods
[1]	Mavenir	Fronthaul (split 7.2)	O	Throughput	O	Only Feasibility verification
[2]	OAI	nFAPI (split 6)	X	X	X	emulator「5G-EMANE」
[3]	No implement	FAPI (split 6)	X	X	X	Sub scheduler in Layer1
This thesis	OAI	nFAPI (split 6)	O	Throughput/ HARQ RTT	O	Delay manager for os/network latency

reference:

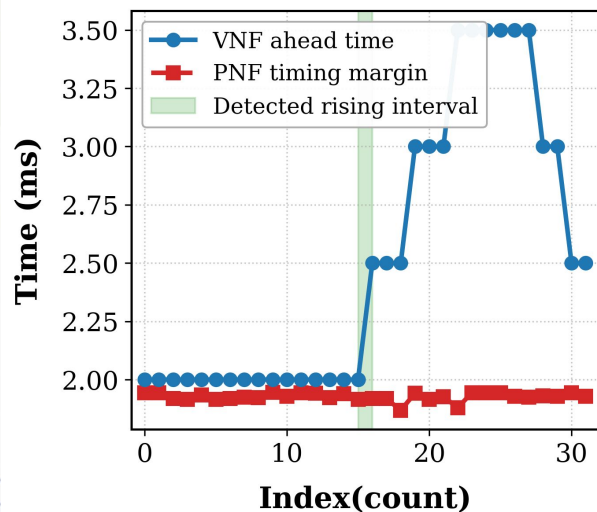
- [1] [F. Mehran and R. MacKenzie, "Experimental evaluation of multi- vendor virtualized ran using non-ideal fronthaul," in 2020 IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications, 2020.](#)
- [2] [B. Ryu, R. Knopp, M. Elkadi, D. Kim, and A. Le, "5g-emane: Scalable open-source real-time 5g new radio network emulator with emane," in MILCOM 2022 - 2022 IEEE Military Communications Conference \(MILCOM\), 2022.](#)
- [3] [S.-Q. Lee and M.-S. Lee, "A method of adaptive low-latency downlink transmission on fapi based 5g nr gnb," in 2022 13th International Conference on Information and Communication Technology Convergence \(ICTC\), 2022.](#)



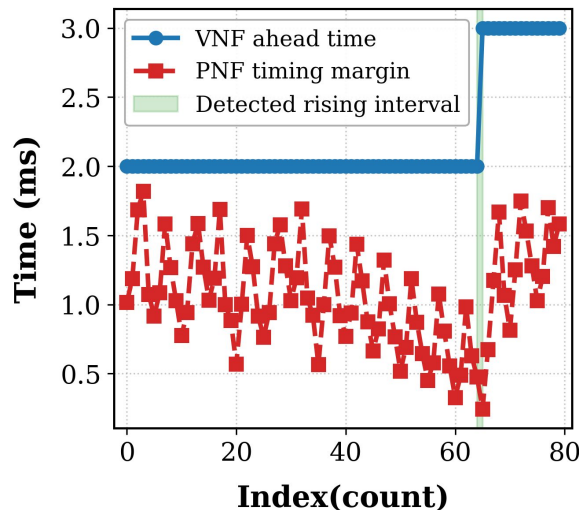
Exp 1: Performance:

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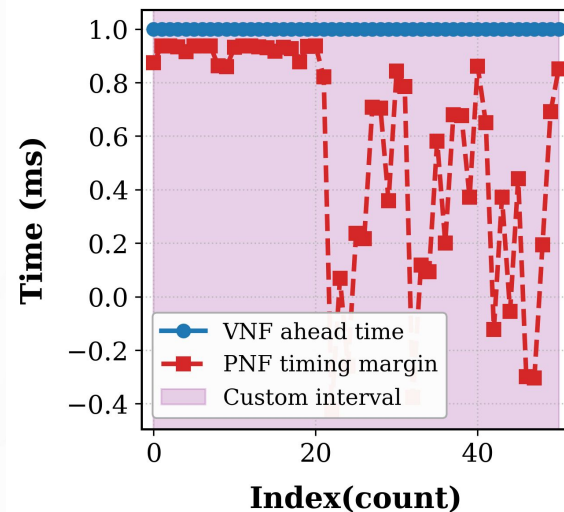
EWMA ($\alpha=1/16, \beta=1/16$)



EWMA ($\alpha=\frac{1}{8}, \beta=\frac{1}{4}$)



EWMA ($\alpha=\frac{1}{2}, \beta=\frac{1}{2}$)



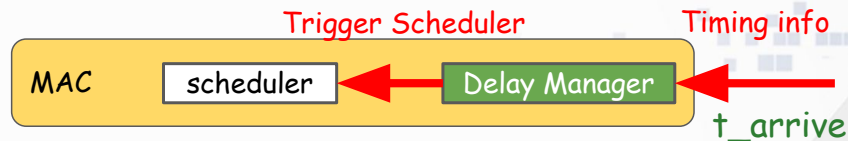
Adopt the parameter settings defined in RFC 6298



Delay Management

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t_{arrive} : PNF report timing info



Delay Management Algorithm

1. Pre-processing

EWMA: α (Stability), β (Response)

2. Anomaly Detection

• IsAnomaly: ($|t_{arrive} - \mu_{delay}| > 3 * dev_{delay}$)
 $\vee (t_{arrive} > slot_duration)$

• Goal: Filter noise via Normal Distribution

3. Adaptive Timing Adjustment

- Anomaly: Increase slot ahead
- Stable: Decrease slot ahead (Optimize RTT)

EWMA Formula & Definitions

Formula:

$$dev_{delay} = (1 - \beta) * dev_{delay} + \beta * |\mu_{delay} - \mu_{delay}'|$$

$$\mu_{delay} = (1 - \alpha) * \mu_{delay} + \alpha * \mu_{delay}'$$

Symbol Definitions:

- μ_{delay} : Smoothed Arrive Time
- μ_{delay}' : Recent measurement arrive time
- dev_{delay} : Round Trip Time Variation

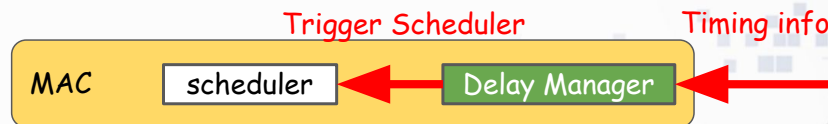
Adopt RFC 6298 settings for EWMA filter:

- **Stability ($\alpha=1/8$):** Avoids sway from packet spikes
- **Response ($\beta=1/4$):** Quick reaction to delay variations



Delay Management

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Delay Management Algorithm

1. Pre-processing

EWMA: α (Stability), β (Response)

2. Anomaly Detection

• IsAnomaly: $(|t_{\text{arrive}} - \mu_{\text{delay}}| > 3 * \text{dev}_{\text{delay}})$
V $(t_{\text{arrive}} > \text{slot_duration})$

• Goal: Filter noise via Normal Distribution

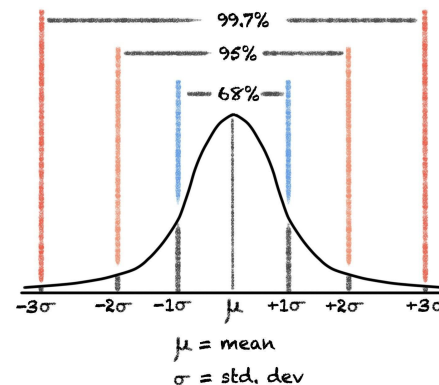
3. Adaptive Timing Adjustment

- Anomaly: Increase slot ahead
- Stable: Decrease slot ahead (Optimize RTT)

Normal Distribution Logic:

3σ covers 99.7% of data. Ensures robust anomaly filtering.

Normal Distribution



Neil Kakkar



Delay Management: Node Sync

Algorithm 1: Adaptive Node Synchronization

Input : t_1, t_2, t_3 (Timestamps from PNF *UL_node_sync* message), t_4 (VNF local receive timestamp)

Output : S_{adj} (Slot adjustment), $\mu_{S_{adj}}$ (Microsecond adjustment)

Notation: *offset* (Instantaneous timing offset), μ_{offset} (EWMA mean offset), dev_{offset} (EWMA mean absolute deviation / Jitter), *Threshold* (Dynamic synchronization threshold), α, β (Smoothing factors, where $\alpha = 1/8, \beta = 1/4$)

/* 1. Calculate Raw Time Differences (with SFN wrap-around protection) */

1 $d_1 \leftarrow t_2 - t_1$

2 $d_2 \leftarrow t_4 - t_3$

/* 2. Estimate Offset and Dynamic Jitter */

3 $offset \leftarrow (d_1 - d_2)/2$

4 $error \leftarrow offset - \mu_{offset}$

5 $\mu_{offset} \leftarrow \mu_{offset} + \alpha \cdot error$

6 $dev_{offset} \leftarrow dev_{offset} + \beta(|error| - dev_{offset})$

7 $Threshold \leftarrow dev_{offset}$

/* 3. Phase Alignment */

8 **if** $|offset| > Threshold$ **then**

9 $S_{adj} \leftarrow \lfloor offset / slot_duration \rfloor$

10 $\mu_{S_{adj}} \leftarrow offset \pmod{slot_duration}$

11 Apply S_{adj} and $\mu_{S_{adj}}$ to adjust local clock

12 **else**

13 State $\leftarrow$ **Synchronized**

14 $S_{adj} \leftarrow 0, \mu_{S_{adj}} \leftarrow 0$

15 **return** $S_{adj}, \mu_{S_{adj}}$



Delay Management: Node Sync

Algorithm 1: Node Synchronization: Adaptive Phase Offset Alignment

Input : t_1, t_2, t_3 (Timestamps from PNF *UL_node_sync* message)

Output : S_{adj} and $\mu_{s_{adj}}$ (Slot and microsecond adjustments)

Notation: t_4 (VNF local receive timestamp), Δt_{offset} (Instantaneous clock phase offset), μ (EWMA mean phase offset), v (Expected phase jitter tracked by EWMA), Tol (Dynamic synchronization threshold), α, β (Smoothing factors, where $\alpha = 1/8, \beta = 1/4$)

1 Phase I: Timestamp Processing and Wrap-around Protection

2 Record local receive timestamp t_4 upon arrival of synchronization message

3 Calculate raw time differences: $d_1 \leftarrow t_2 - t_1$ and $d_2 \leftarrow t_4 - t_3$

4 Apply 10.24s wrap-around protection to d_1 and d_2 based on SFN cycle constraints

5 Phase II: Offset and Adaptive Threshold Estimation

6 Calculate instantaneous timing offset: $\Delta t_{offset} \leftarrow (d_1 - d_2)/2$

7 Calculate phase offset error: $e \leftarrow \Delta t_{offset} - \mu$

8 Update EWMA mean phase offset: $\mu \leftarrow \mu + \alpha \cdot e$

9 Update expected phase jitter: $v \leftarrow v + \beta(|e| - v)$

10 Set dynamic synchronization threshold: $Tol \leftarrow v$



Delay Management: Node Sync

11 Phase III: Phase Alignment and Convergence

12 **if** $|\Delta t_{offset}| > Tol$ **then**

13 Compute discrete slot shift: $S_{adj} \leftarrow \lfloor \Delta t_{offset} / \text{slot duration} \rfloor$

14 Compute residual microsecond shift: $\mu S_{adj} \leftarrow \Delta t_{offset} \pmod{\text{slot duration}}$

15 Apply S_{adj} and μS_{adj} to align VNF master clock schedule

16 **else if** $|\Delta t_{offset}| \leq Tol$ **then**

17 Transition to **Synchronization Achieved** state

18 Lock phase alignment and monitor for long-term drift

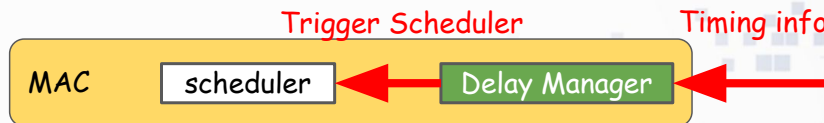
19 $S_{adj} \leftarrow 0, \mu S_{adj} \leftarrow 0$

20 **return** $S_{adj}, \mu S_{adj}$



Delay Management

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Algorithm 2: Trigger Timing Control

Input : t_{arrive} (Timing info from PNF)

Output : S_{next} (Trigger timing for the next MAC scheduling)

Notation: S_{curr} (Current trigger timing), μ_{delay} (EWMA mean delay), dev_{delay} (EWMA mean deviation), $count_{stable}$ (Consecutive stable arrivals counter), N_{stable} (Adaptive stability threshold, dynamically scaled by dev_{delay}), α, β (Smoothing factors, where $\alpha = 1/8, \beta = 1/4$)

1 Phase I: Delay Statistics Update

2 $error \leftarrow t_{arrive} - \mu_{delay}$

3 $\mu_{delay} \leftarrow \mu_{delay} + \alpha \cdot error$

4 $dev_{delay} \leftarrow dev_{delay} + \beta(|error| - dev_{delay})$

5 Phase II: Anomaly Detection

6 $IsAnomaly \leftarrow (|error| > 3 \cdot dev_{delay}) \vee (t_{arrive} > slot_duration)$

7 Phase III: Timing Adjustment

8 **if** $IsAnomaly$ **then**

9 | $S_{next} \leftarrow S_{curr} + 1$

10 | $count_{stable} \leftarrow 0$

11 **else if** $t_{arrive} \leq Safe_Margin$ **then**

12 | $count_{stable} \leftarrow count_{stable} + 1$

13 | **if** $count_{stable} \geq N_{stable}$ **then**

14 | $S_{next} \leftarrow S_{curr} - 1$

15 | $count_{stable} \leftarrow 0$

16 **else**

17 | $S_{next} \leftarrow S_{curr}$

18 **else**

19 | $S_{next} \leftarrow S_{curr}$

20 | $count_{stable} \leftarrow 0$

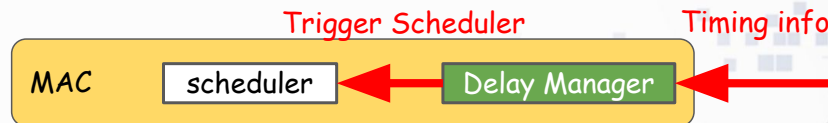
21 Limit S_{next} within the allowable nFAP timing window

22 **return** S_{next}



Delay Management

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Algorithm 2: Delay Manager: Statistical Variance-Based Control (EWMA)

Input : t_{late} (Timing info including latest arriving timing offset from PNF)

Output : S_{next} (Trigger Scheduler timing for the next MAC scheduler)

Notation: S_{curr} (Current trigger timing), μ (EWMA mean timing offset), v (EWMA mean absolute deviation), C_{stable} (Counter for consecutive stable packet arrivals), P (Dynamic stability threshold for latency reduction), α, β (Smoothing factors, where $\alpha = 1/8, \beta = 1/4$)

1 Phase I: EWMA Estimation (Prevent Ping-Pong Effect)

2 Calculate timing error: $e \leftarrow t_{late} - \mu$

3 Update mean timing offset: $\mu \leftarrow \mu + \alpha \cdot e$

4 Update mean absolute deviation: $v \leftarrow v + \beta(|e| - v)$

5 Phase II: Boundary & Anomaly Detection

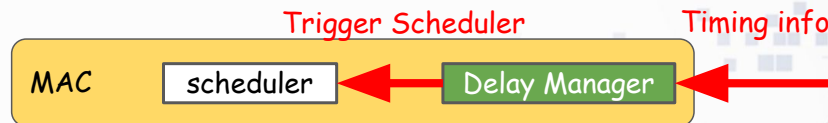
6 Calculate safety boundary B_{safe} based on S_{curr} and system timing window

7 Identify anomaly: $IsAnomaly \leftarrow |e| > 3v$

8 Set $Panic_Mode$ **true** if $IsAnomaly$ occurs or t_{late} exceeds one slot duration



Delay Management



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```

9 Phase III: Fast-up & Slow Decay
10 if Panic Mode is triggered then
11   Advance trigger timing:  $S_{next} \leftarrow S_{curr} + 1$ 
12   Increase stability threshold  $P$  and reset  $C_{stable}$  to zero
13 else if  $t_{late}$  is within the absolute safety boundary  $-B_{safe}$  then
14   Increment  $C_{stable}$  to track continuous stable arrivals
15   if  $P$  is above baseline then
16     Decay stability threshold  $P$  by one unit
17   if  $C_{stable} > P$  then
18     Retard trigger timing:  $S_{next} \leftarrow S_{curr} - 1$  and reset  $C_{stable}$ 
19   else
20     Maintain current timing:  $S_{next} \leftarrow S_{curr}$ 
21 else
22   Maintain current timing:  $S_{next} \leftarrow S_{curr}$  and reset  $C_{stable}$ 
23 Limit  $S_{next}$  within the allowable system timing window
24 return  $S_{next}$ 
  
```



Downlink Data Delivery Status (DDDS)

5.4.2 Downlink Data Delivery Status

5.4.2.1 Successful operation

The purpose of the Downlink Data Delivery Status procedure is to provide feedback from the corresponding node to the node hosting the NR PDCP entity to allow the node hosting the NR PDCP entity to control the downlink user data flow via the corresponding node for the respective data radio bearer. The corresponding node may also transfer uplink user data for the concerned data radio bearer to the node hosting the NR PDCP entity together with a DL DATA DELIVERY STATUS frame within the same GTP-U PDU.

The Downlink Data Delivery Status procedure is also used to provide feedback from the corresponding node to the node hosting the NR PDCP entity to allow the node hosting the NR PDCP entity to control the successful delivery of DL control data to the corresponding node.

When the corresponding node decides to trigger the feedback for Downlink Data Delivery procedure it shall report as specified in section 5.2:

- a) in case of RLC AM, the highest NR PDCP PDU sequence number successfully delivered in sequence to the UE among those NR PDCP PDUs received from the node hosting the NR PDCP entity i.e. excludes those retransmission NR PDCP PDUs;

NOTE 1: If the NR user plane protocol instance is associated a single RLC-AM entity for an MRB, specification text in bullet a) is applicable.

For all other cases, if the NR user plane protocol instance is associated with an MRB configured with at least one RLC AM entity and RLC-UM, the highest successfully delivered NR PDCP sequence number indicates the combined feedback of the highest NR PDCP sequence number successfully delivered in sequence to all the involved UEs for which the RLC AM entities have been configured, no retransmissions are performed, and the highest NR PDCP sequence number transmitted to the lower layers for PTM.

- b) the desired buffer size in bytes for the concerned data radio bearer or for the MRB;
- c) optionally, the desired data rate in bytes associated with a specific data radio bearer configured for the UE or for the MRB;
- d) the NR-U packets that were declared as being "lost" by the corresponding node and have not yet been reported to the node hosting the NR PDCP entity within the DL DATA DELIVERY STATUS frame;
- e) if retransmission NR PDCP PDUs have been delivered, the NR PDCP PDU sequence number associated with the highest NR-U sequence number among the retransmission NR PDCP PDUs successfully delivered to the UE in sequence of NR-U sequence number;
- f) if retransmission NR PDCP PDUs have been transmitted to the lower layers, the NR PDCP PDU sequence number associated with the highest NR-U sequence number among the retransmission NR PDCP PDUs transmitted to the lower layers in sequence of NR-U sequence number;